# 7.3 Proving That a Quadrilateral Is a Parallelogram

Name _____ Period ____ Date _____

Geometry 7.3 · Student sheet

**I can…**

- name the theorem that proves a quadrilateral is a parallelogram.
- find values of variables that make a quadrilateral a parallelogram.
- use slope, distance, or midpoints to show a quadrilateral in the coordinate plane is a parallelogram.

**Vocabulary** · fill in the missing word

**Converse** — the statement made by switching the _____ and the conclusion

**Parallelogram Opposite Sides Converse** — both pairs of opposite sides _____ → parallelogram

**Parallelogram Opposite Angles Converse** — both pairs of opposite angles _____ → parallelogram

**Opposite Sides Parallel and Congruent Theorem** — one _____ of opposite sides both congruent and parallel → parallelogram

**Parallelogram Diagonals Converse** — diagonals _____ each other → parallelogram

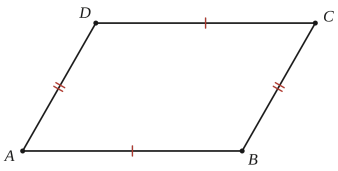
## Part 1 · Build it: does it have to be a parallelogram?

Connect two strips of one length and two of another, with equal strips **opposite** each other. Can you make a quadrilateral that is NOT a parallelogram?

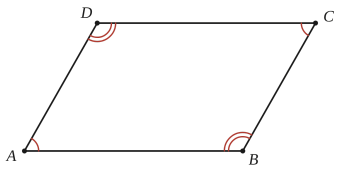
Now put the equal strips **next to** each other. What shape do you get?

**Is it a parallelogram?** Write yes or not enough information, and the reason.

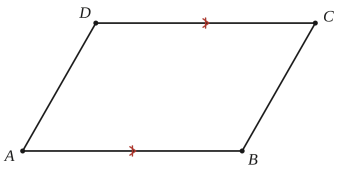
**1.**



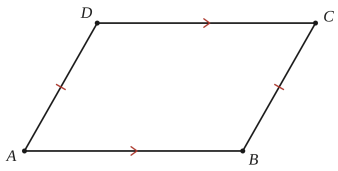

**2.**




**3.**




**4.**




## Part 2 · Notes and examples (fill in each note, then work the example)

**Definition of a parallelogram:** A quadrilateral is a parallelogram if both pairs of opposite sides are _____ (definition).

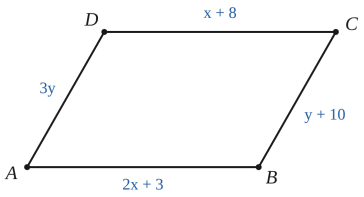
**Parallelogram Opposite Sides Converse:** A quadrilateral is a parallelogram if both pairs of opposite sides are _____.

**Parallelogram Opposite Angles Converse:** A quadrilateral is a parallelogram if both pairs of opposite angles are _____.

### Example 1

For what values of x and y is ABCD a parallelogram?

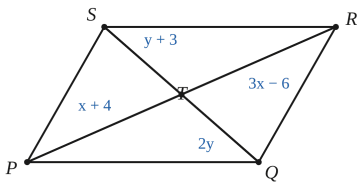
AB = 2x + 3, DC = x + 8, AD = 3y, BC = y + 10




**Parallelogram Diagonals Converse:** A quadrilateral is a parallelogram if the diagonals _____ each other.

### Example 2

The diagonals of quadrilateral PQRS meet at T. PT = x + 4, TR = 3x − 6, QT = 2y, and TS = y + 3. For what values of x and y is PQRS a parallelogram?




**Coordinates:** use _____ (parallel), _____ (congruent), or _____ of the diagonals (bisect).

### Example 3

Show that A(−3, 3), B(1, 4), C(3, 0), and D(−1, −1) are the vertices of a parallelogram.

**Opposite Sides Parallel and Congruent Theorem:** A quadrilateral is a parallelogram if one pair of opposite sides is _____ and _____.

### Example 4

In quadrilateral WXYZ, $\overline{WX} \cong \overline{YZ}$ and $\overline{WZ} \parallel \overline{XY}$. Is WXYZ a parallelogram?

## Part 3 · Partner practice

### Partner 1

In quadrilateral ABCD, $m\angle A = (4x - 10)°$ and $m\angle C = (2x + 30)°$. For what value of x could ABCD be a parallelogram (with $m\angle B = m\angle D$)?

### Partner 2

Is EFGH with E(0, 0), F(4, 1), G(5, 4), and H(1, 3) a parallelogram? Use midpoints.